Exp. No: 18	OPTICAL SPECTRUM AND OSNR ANALYSIS USING OPTISYSTEM
Date:	

OBJECTIVES:

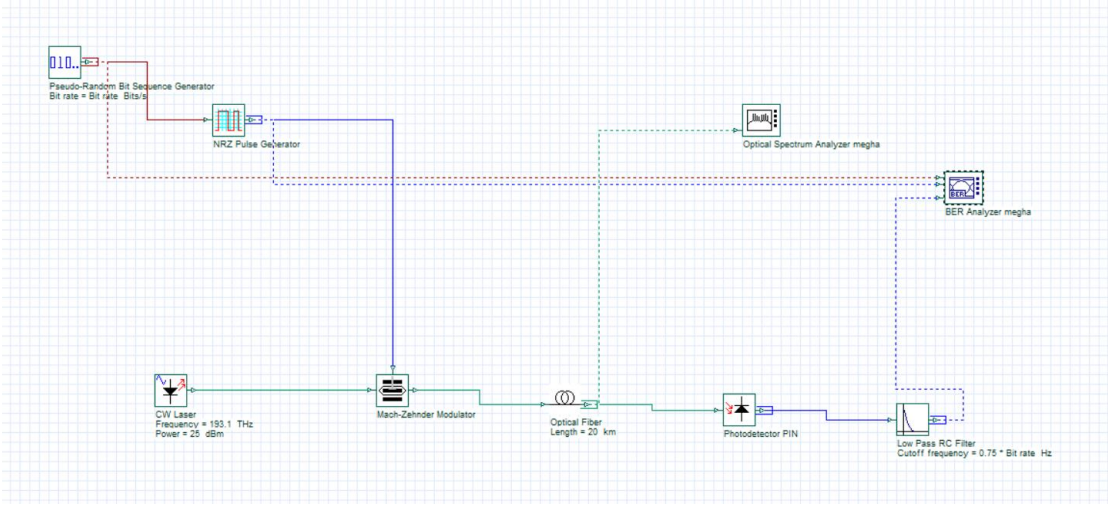
1. Design an optical communication system using OptiSystem.
2. Study how source power affects optical spectral intensity.
3. Analyze how fiber length affects attenuation, dispersion, and nonlinear effects.
4. Study the combined effect of source power and fiber length.

EXPERIMENT PROCEDURE:

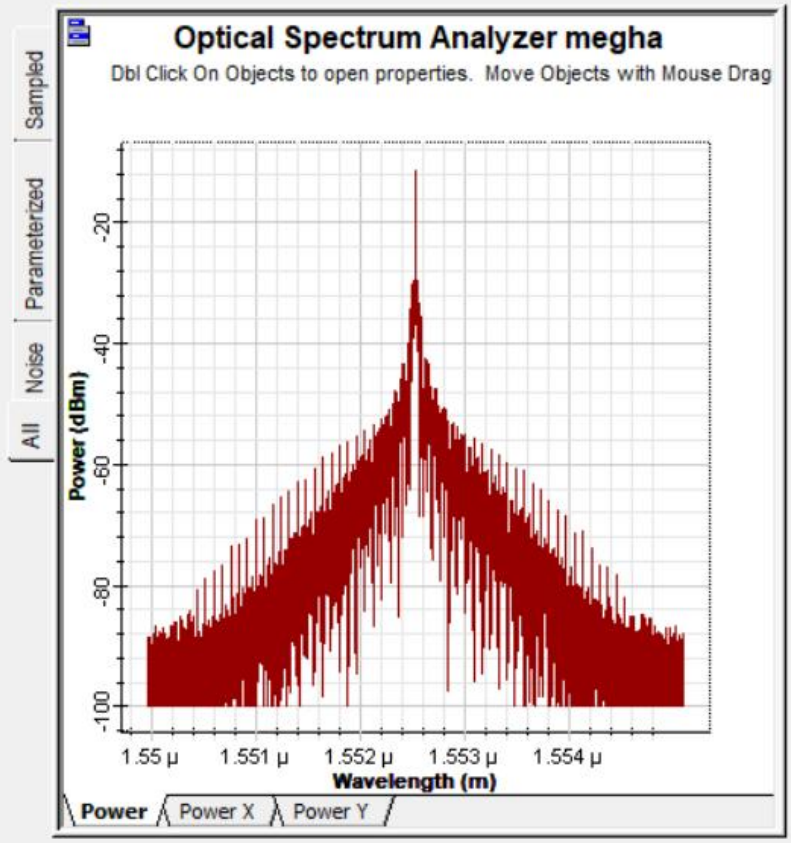
1. Design the optical link in OptiSystem using laser, NRZ generator, MZ modulator, fiber
and PIN photodiode
2. Connect the Optical Spectrum Analyzer to observe the optical spectrum.
3. Set wavelength to 1550 nm
4. Vary source power and fiber length according to the test cases
5. Run the simulation and record spectral power, spectral width and received power
6. Compare the results for different power and fiber-length values

INPUT PARAMETERS TABLE:

PARAMETERS	VALUES
Wavelength	1550 nm
Modulation	NRZ
Fiber Type	SMF
Attenuation	0.2 db/km
Dispersion	16 ps/nm/km
Source Power	Variable
Fiber Length	variable



Optical Spectrum Analyzer



Signal Index: 0

Auto Set

Wavelength

Units: m

Automatic range

Center: 1.55253e-006 m

Start: 1.5497e-006 m

Stop: 1.55536e-006 m

Amplitude

Units: dBm

Automatic range

Max: -6.90266 dBm

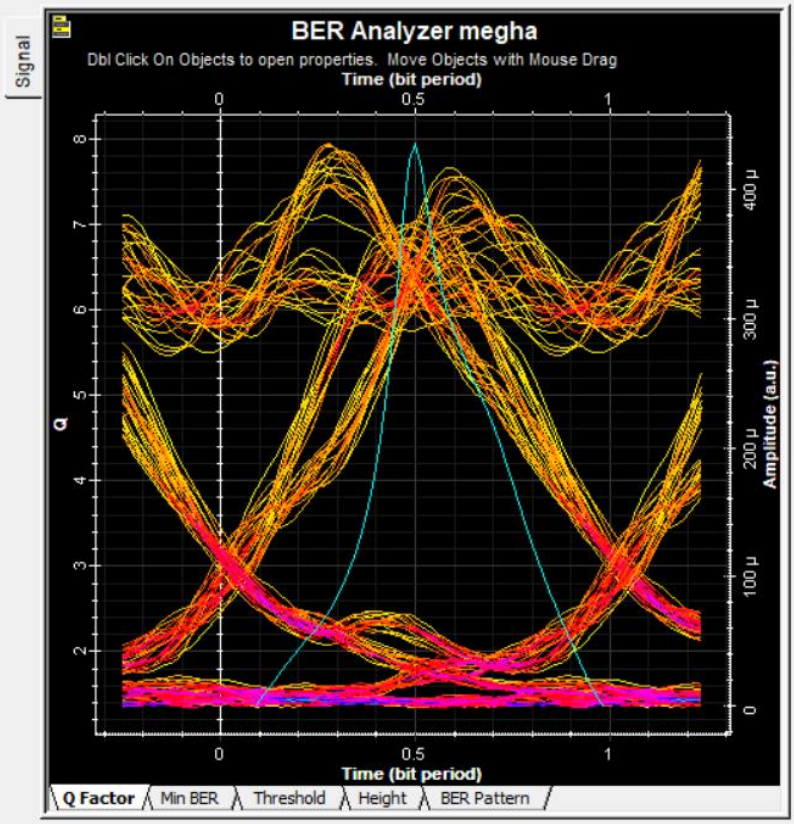
Min: -104.433 dBm

Resolution Bandwidth

Res: 0.01 nm

Invert Colors

BER Analyzer



Signal Index: 0

Auto Set

Show Eye Diagram

Analysis

Max. Q Factor	7.93841
Min. BER	1.02389e-015
Eye Height	0.000192412
Threshold	0.000183111
Decision Inst.	0.5

Invert Colors

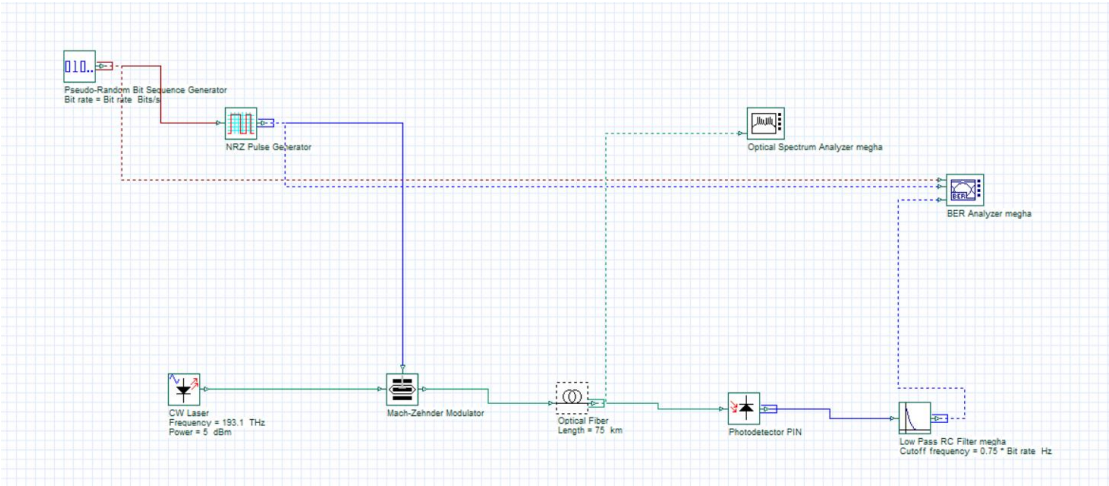
Color Grade

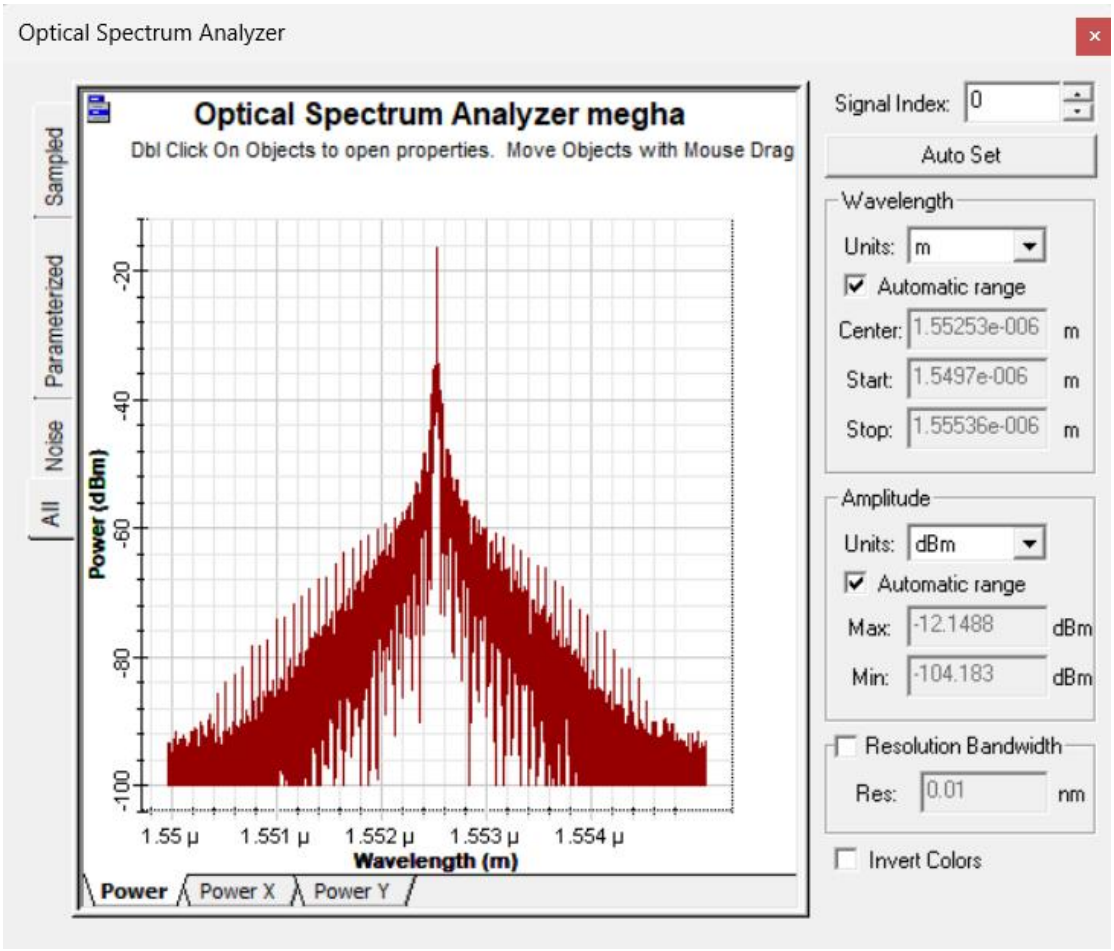
Patterns

Calculate Patterns

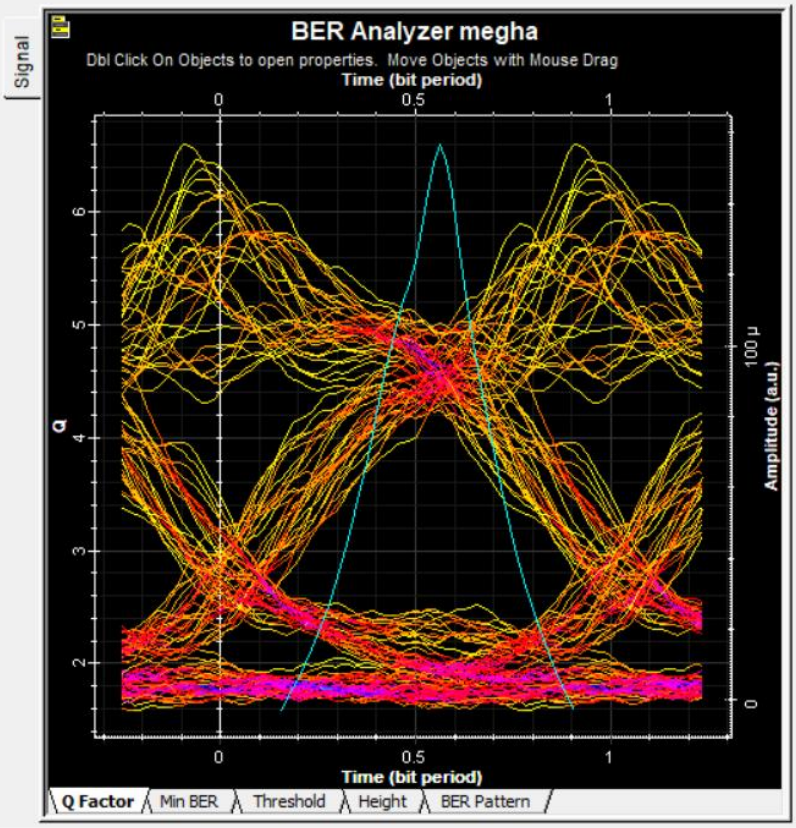
Patterns

Pattern 1	1e-012
Pattern 2	1e-011
Pattern 3	1e-010
Pattern 4	1e-009
Pattern 5	1e-008


Pseudo-Random Bit Sequence Generator
Bit rate = Bit rate Bits/s
NRZ Pulse Generator
Optical Spectrum Analyzer megha
BER Analyzer megha
CW Laser
Frequency = 193.1 THz
Power = 5 dBm
Mach-Zehnder Modulator
Optical Fiber
Length = 75 km
Photodetector PIN
Low Pass RC Filter megha
Cutoff frequency = 0.75 * Bit rate Hz


Optical Spectrum Analyzer
Optical Spectrum Analyzer megha
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Sampled
Parameterized
Noise
All
Power (dBm)
-20
-40
-60
-80
-100
1.55 µ
1.551 µ
1.552 µ
1.553 µ
1.554 µ
Wavelength (m)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: -12.1488 dBm
Min: -104.183 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors

BER Analyzer



Signal Index: 0

Auto Set

Show Eye Diagram

Analysis

Max. Q Factor	6.6025
Min. BER	2.0209e-011
Eye Height	4.59723e-005
Threshold	5.26014e-005
Decision Inst.	0.5625

Invert Colors

Color Grade

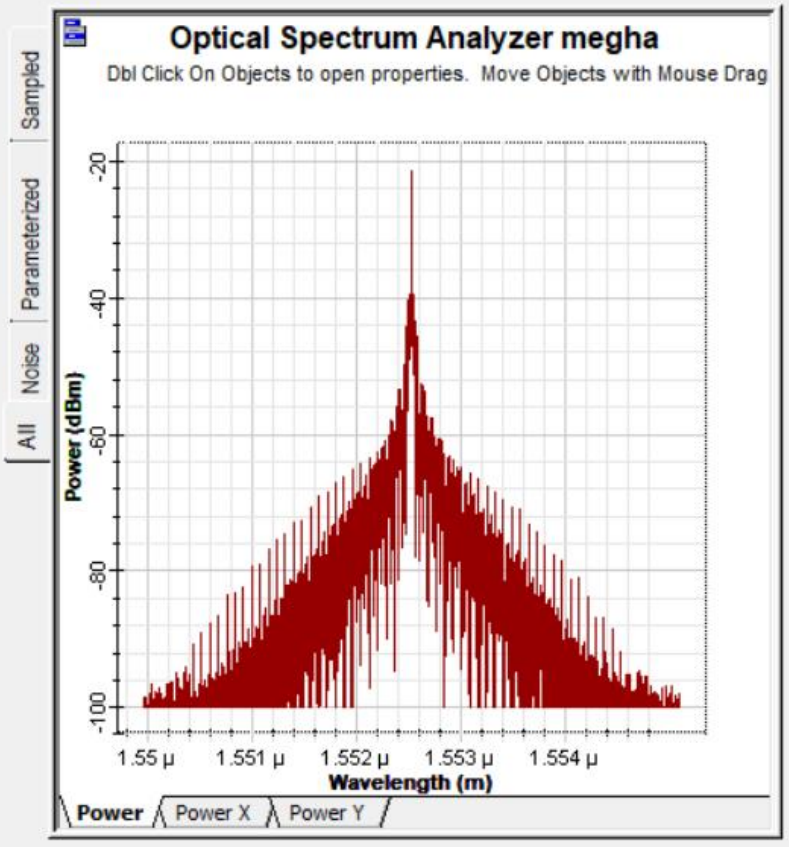
Patterns

Calculate Patterns

Patterns

Pattern 1	1e-012
Pattern 2	1e-011
Pattern 3	1e-010
Pattern 4	1e-009
Pattern 5	1e-008

Optical Spectrum Analyzer



Signal Index: 0

Auto Set

Wavelength

Units: m

Automatic range

Center: 1.55253e-006 m

Start: 1.5497e-006 m

Stop: 1.55536e-006 m

Amplitude

Units: dBm

Automatic range

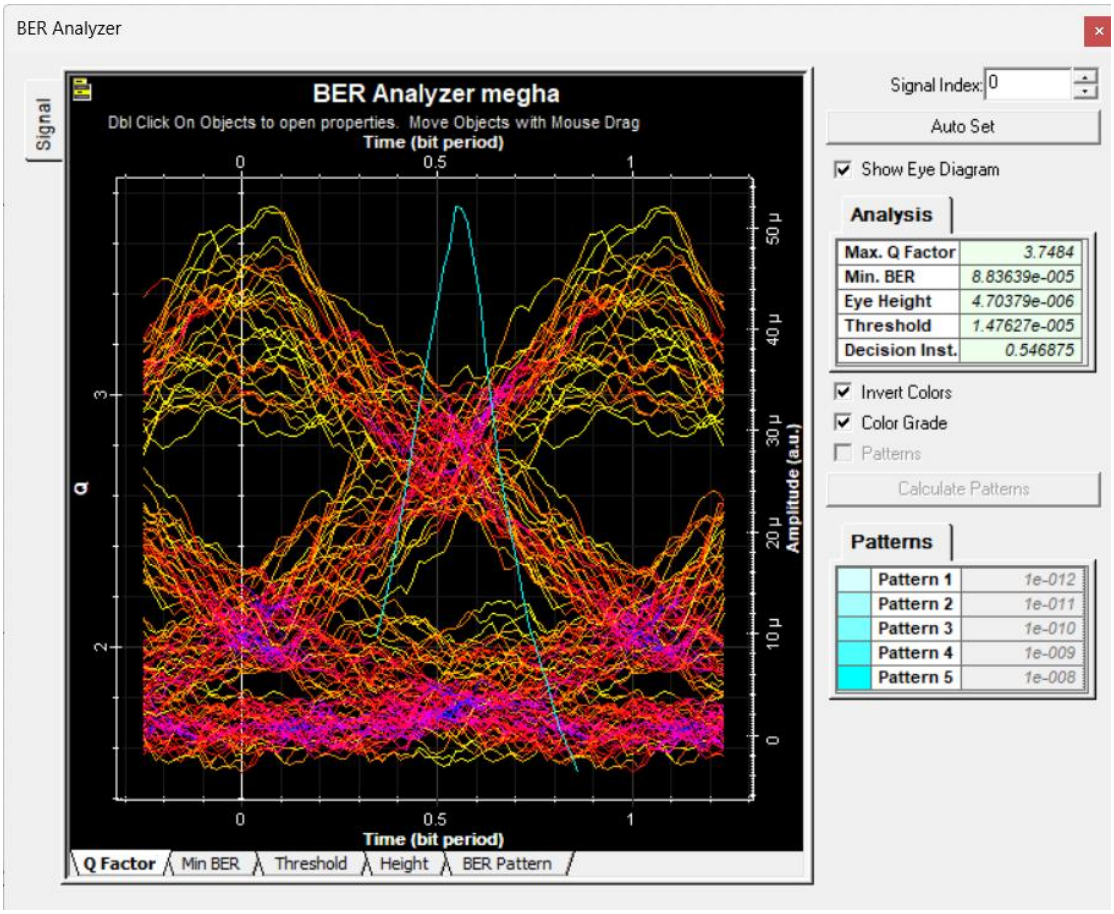
Max: -17.3976 dBm

Min: -103.933 dBm

Resolution Bandwidth

Res: 0.01 nm

Invert Colors



OUTPUT PARAMETERS TABLE:

Power	Distance	BER	Q-Factor
5 dbm	50km	1.58374e-014	7.59093
	75km	7.06805e-011	6.414
	100km	3.3262	0.000440197
	150km	1	0
10 dbm	50km	1.83955e-013	7.26649
	75km	1.46145e-014	7.60095
	125km	0.00771314	2.42192
	150km	1	0
15 dbm	50km	1.96692e-010	6.23783
	75km	4.8004e-020	9.09293
	100km	1.38127e-007	5.12115
	150km	0.0281456	1.88966
	200km	1	0
	250km	1	0

Student Reflection:

Through this OptiSystem simulation, I understood how source power and optical fiber length affect the performance of an optical communication system. I learned how to observe the optical spectrum and analyze the Bit Error Rate (BER) and Q-factor under different operating conditions. The experiment helped me understand the effects of fiber attenuation, dispersion, and nonlinearities on signal quality and transmission distance. I also learned how to identify suitable source power and fiber length values for achieving better optical communication performance.

Result:

An optical communication system was successfully designed and simulated using OptiSystem at a wavelength of 1550 nm. The effects of source power and fiber length on BER and Q-factor were analyzed. The results showed that increasing fiber length generally degrades signal quality due to attenuation and dispersion, while higher source power can improve performance under certain conditions. The best recorded performance was obtained at 15 dBm source power and 75 km fiber length, with a BER of 4.8004×10^{-20} and a Q-factor of 9.09293.